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Bridging the gap of brain and experience – Converging Neurophenomenology with Spatiotemporal Neuroscience

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ABSTRACT

Neuroscience faces the challenge of connecting brain and mind, with the mind manifesting in first-person experience while the brain's neural activity can only be investigated in third-person perspective. To connect neural and mental states, Neurophenomenology provides a methodological toolkit for systematically linking first-person subjective experience with third-person objective observations of the brain's neural activity. However, beyond providing a systematic methodological strategy ('disciplined circularity'), it leaves open how neural activity and subjective experience are related among themselves, independent of our methodological strategy. The recently introduced Spatiotemporal Neuroscience suggests that neural activity and subjective experience share a commonly underlying feature as their "common currency", notably analogous spatiotemporal dynamics. Can Spatiotemporal Neuroscience inform Neurophenomenology to allow for a deeper and more substantiative connection of first-person experience and third-person neural activity? The goal of our paper is to show how Spatiotemporal Neuroscience and Neurophenomenology can be converged and integrated with each other to gain better understanding of the brain-mind connection. We describe their convergence on theoretical grounds which, subsequently, is illustrated by empirical examples like self, meditation, and depression. In conclusion, we propose that the integration of Neurophenomenology and Spatiotemporal Neuroscience can provide complementary insights, enrich both fields, allows for deeper understanding of brain-mind connection, and opens the door for developing novel methodological approaches in their empirical investigation.

1. Introduction

How are brain and mind connected with each other? While this question has been debated traditionally in philosophy (e.g., Descartes, 1641; Kant, 1781; Nagel, 1974), neuroscience has taken major empirical steps in investigating the neural basis of mental features like consciousness (Northoff and Lamme, 2020; Seth and Bayne, 2022; Storm et al., 2024), selfhood (Northoff et al., 2024, 2016), and thought processes (Christoff et al., 2016; Hua et al., 2022). In parallel, neuroscience has also sought to uncover the neurocomputational mechanisms that underlie brain processes and their neural activity. Several theoretical models, including predictive coding and the free energy principle (Friston, 2010), thermodynamics (Kringelbach et al., 2024), the inside-out model (Buzsáki, 2019), and the intrinsic model (Raichle, 2009, 2010), attempt to explain these fundamental mechanisms, such as recurrent connections, excitation-inhibition balance, and other key processes. Despite these major advances, the nature of the relationship between neural processes and mental states remains yet elusive, though.

One key issue is how to connect subjective experience with the objective observation of the brain's neural activity. This is a major methodological problem: the brain's neuronal processes can only be observed from a third-person perspective (3PP), while subjective experience is accessible exclusively from a first-person perspective (1PP). To establish a meaningful connection between brain and experience, it is thus necessary to bridge these two perspectives. Neurophenomenology seeks to accomplish precisely this: it offers a systematic methodological approach for bridging 1PP and 3PP, where each perspective informs the other through what has been termed "disciplined circularity" (Varela, 1996; Lutz, 2002; Lutz et al., 2024; Lutz and Thompson, 2003; Berkovich-Ohana et al., 2020; Kyzar and Denfield, 2023).

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Despite the major progress of Neurophenomenology, it does not yet fully explain the intrinsic connection between neural processes in the brain and subjective experience, beyond the integration of 1PP and 3PP (and thus independent of our first-third person investigation and its methodological strategy). This gap is addressed by the recently introduced framework of Spatiotemporal Neuroscience (Northoff, 2020, 2024; Northoff et al., 2024), which, roughly, proposes that brain activity and experience share a common feature: their spatiotemporal dynamics, serving as a "common currency."

Furthermore, Spatiotemporal Neuroscience comes with a novel methodological approach, namely non-reductive neurophilosophy (NRNP; Northoff, 2014, 2022a, 2022b) (as distinguished from its reductive sibling, namely reductive neurophilosophy (Churchland 1986, 2002). Through a process called "concept-fact iterativity," this non-reductive method moves beyond simply using both first-person (e.g., phenomenology) and third-person (e.g., neural analyses) approaches in parallel. Instead, it seeks to identify the underling features that are shared by both third-person neural and first-person experiential processes, i.e., their common currency. By methodologically moving back and forth in an iterative way between the third-person data/facts related to neural features and the concepts describing first-person subjective experience, concept-fact iterativity allows for a more direct and intrinsic connection of brain and mind (Northoff, 2014, 2022a, 2022b).

Can we converge the methodological strategy of Neurophenomenology ('disciplined circularity') with the shared temporal-spatial dynamics of brain-mind as proposed in Spatiotemporal Neuroscience and the concept-fact iterativity of NRNP? Proposing such convergence with the aim of showing the deeper and more substantial connection of brain and experience is the goal of our paper. For this purpose, we will first introduce both Neurophenomenology and Spatiotemporal Neuroscience, second, discuss their theoretical convergence, and third, illustrate that by empirical examples.

2. Part I: what are Neurophenomenology and Spatiotemporal Neuroscience? Brief introduction

2.1. Neurophenomenological approach: methodological strategies for connecting first- and third-person perspectives

Neurophenomenology was initially introduced by Francisco Varela as a methodological approach to the scientific investigation of consciousness (Varela, 1996; Varela and Shear, 1999). It emphasizes the careful examination of experience using rigorous first-person and third-person methodologies inspired by phenomenology and contemplative practices. This allows for the systematic investigation of experiential contents ('What' appears to the experiencer) and their structure ('How' it appears) (e.g., sense of time, self, subject-object duality). These first-person accounts of experience can then guide and inform the interpretation of neurophysiological processes relevant to consciousness in such a way that first-person data and third-person empirical data mutually constrain each other through what is described as 'disciplined circularity'.

Crucially, 1PP and 3PP are mutually exclusive: 1PP does not provide direct access to the brain, while 3PP cannot capture subjective experience. This distinction underscores the inherent difference between these two perspectives. Neurophenomenology seeks to bridge this gap between 1PP and 3PP by identifying the neural correlates of specific phenomenal features of experience. In this vein, it can be seen to provide methodological strategies designed to explore how 1PP (subjective experience) and 3PP (neural activity) can be connected in a systematic way in our investigation of brain-mind.

In their recent review, Lutz et al. (2024) presented a comprehensive framework comprising five (neuro)phenomenological methodological approaches that progressively integrate subjective experience with neuroscientific data. The first approach, *phenomenological inquiries*, employs first-person methods like micro-phenomenological interviews to capture the nuanced nature of subjective experiences, as it is referred to as "thick phenomenology." Building on this foundation, a second approach called *experimental or clinical phenomenology* integrates phenomenological methods with quantitative tools, such as psychometric scales or experience-sampling, to measure statistical patterns across subjective reports, resulting in "thin phenomenology." This approach enables a more systematic and data-driven investigation of phenomenological features. The third approach, *experimental neurophenomenology*, further merges first-person data with third-person data, including behavioral, neural and physiological measurements. Here, trained participants are used to establish mutual constraints between their subjective experiences and the own neural or behavioral correlates.

Moving further, *computational phenomenology* focuses on modeling phenomenological invariants through computational techniques, aiming to create explanatory and predictive models that bridge and connect first-person and third-person data (Hedrih and Hedrih, 2016; Beckmann et al., 2023). The most advanced approach in this framework is *deep computational neurophenomenology*. It combines first-person phenomenology, computational modeling, and neurobiological data to develop generative models that explain and predict both neural and experiential processes at the same time. This approach involves a disciplined interplay between first- and third-person data, facilitated by formal computational frameworks like active inference (Sandved-Smith et al., 2021; Bogotá and Djebbara, 2023; Ramstead et al., 2022) (see also Box 1).

The epistemic value of these methodologies has been discussed in clinical contexts and demonstrated through various empirical studies (Kyzar and Denfield, 2023). These investigations have revealed novel neurodynamic correlates of attention (Lutz, 2002), provided insights into the plasticity of false memory (Petitmengin et al., 2013; Černe and Kordeš, 2023), highlighted the volitional plasticity of self-awareness and its boundaries (Dor-Ziderman et al., 2013, 2016), and successfully integrated neurofeedback with mindfulness meditation practices (Garrison et al., 2013a, 2013b; van Lutterveld et al., 2017a). Collectively, these methodological approaches offer promising pathways for bridging the gap between subjective experience and objective neuroscientific data, potentially leading to a more comprehensive and nuanced understanding of brain-mind connection including both consciousness and cognition Fig. 1a.

2.2. Spatiotemporal approach I – neural activity: the brain exhibits inner time and space

What about the spatiotemporal approach? The focus here is on characterizing both brain and experience by themselves independent of their investigation by us. The brain is characterized by neural dynamics, the changing patterns of neural activity, that is, its dynamics (Northoff, 2024) which can only be observed from a 3PP. On the other hand, subjective experience consists of phenomenal or mental states that can be accessed only in experience from a 1PP. This mutual exclusivity of brain and experience imposes constraints on how we can link them in a more direct way. Because each can only be accessed through its specific mutually exclusive perspective, it remains challenging to uncover how brain and experience might be directly related by themselves independent of our methods in both first- and third-person perspective.

To mitigate this challenge, we may need to find a more direct connection between brain and experience, independent of the methodological limitations imposed by these perspectives. This is where the spatiotemporal approach becomes relevant. The spatiotemporal approach posits that brain activity and subjective experience share common features that provide their direct connection: these shared features are supposed to consist in space and time. Both brain and experience are characterized by their own inner, internal or intrinsic temporal and spatial dynamics, distinct from the outer, external or extrinsic time and space of the physical world (Kant, 1781; Northoff, 2012; Northoff et al., 2020; Kent and Wittmann, 2021).

The brain's "inner time" and "inner space" are reflected in its

Box 1
– Dynamic Computational Neurophenomenology.

Neurophenomenological computational models, like those discussed by Lutz et al. (2024), aim to bridge the gap between subjective experience and brain activity through formal, computational frameworks. In particular, "deep computational neurophenomenology" proposes that deep computational processes can provide "generative passages between 1PP and 3PP" (Lutz et al., 2024). These models often use the active inference approach, which conceptualizes mental actions such as attention and meta-awareness as part of a dynamic, hierarchical process of prediction and error correction. In this framework, the brain constantly generates predictions about its internal and external environment, updating them based on incoming sensory data. This allows for a formalized exploration of how we regulate our thoughts and attentional states in real-time, linking first-person subjective experience with third-person neural data (Sandved-Smith et al., 2021; Bogotá and Djebbara, 2023; Ramstead et al., 2022).

We propose that the notion of "deep generative passages" can be further specified and substantiated by focusing on timescales and, more broadly, temporo-spatial dynamics that are shared by 1PP and 3PP. Methods recently employed by Çatal et al. (2024, 2025), and Keskin and Northoff (2024) offer alternative or complementary techniques that could enrich this framework. For example, Çatal et al. (2024), using a neural mass model, showed how intrinsic neural timescales (INTs), which describe how the brain processes information across different time windows, modulate brain activity and behavior. This model could be applied in neurophenomenological research by adding a layer of temporal dynamics that captures how the brain's internal rhythms relate to moment-to-moment 1PP experiences.

Similarly, Çatal et al. (2025) used the Jansen-Rit neural mass model to examine how brain responses to external events, such as through event-related fields (ERFs), can be understood in relation to intrinsic brain dynamics like INT. This method, combined with active inference, could provide a more comprehensive view of how both internal states and external stimuli shape cognitive and emotional responses, helping to model how brain and experiential dynamics are characterized when responding to dynamic environments while maintaining their ongoing predictions. Keskin and Northoff (2024), using Wilson-Cowan neural mass model, provided insight into the topographic aspects of brain function, particularly how regions related to the self and non-self exhibit different patterns of myelination and neural dynamics. By integrating this approach, neurophenomenological models could account for how the brain's topography influences functional differentiation, offering a more complete picture of how subjective experiences of selfhood and identity are reflected in the brain's architecture.

Together, these methods from Çatal et al. (2024, 2025) and Keskin and Northoff (2024) could complement or extend the active inference model employed in Neurophenomenology, allowing for a richer exploration of the temporal, structural, and dynamic properties of brain and experience. Integrating these techniques into a neurophenomenological framework would offer deeper insights into how subjective experience is linked to brain processes, going beyond prediction-based models to include the brain's intrinsic timescales and their topographic organization.

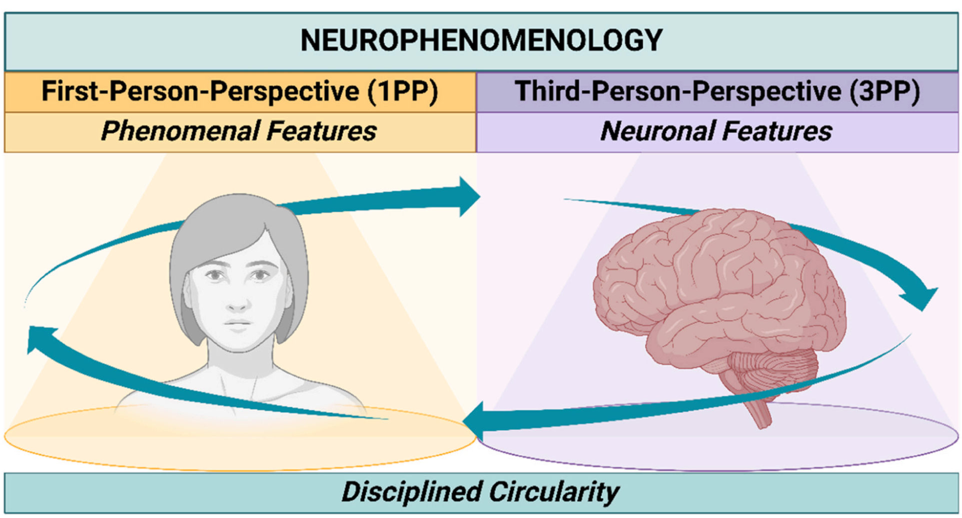

Fig. 1a. The neurophenomenological approach, characterized by disciplined circularity between 1PP and 3PP.

intrinsic topography and neural dynamics. For example, the brain's neural activity exhibits continuously changing patterns over time, known as neural dynamics (Fig. 1b; Northoff, 2024; Cocchi et al., 2017). Importantly, these changes are not random but follow a structured temporal pattern. These dynamic patterns can be measured, for instance, through neural variability (Uddin, 2020; Garrett et al., 2011, 2013, 2018, 2020; Baracchini et al., 2021; Waschke et al., 2021a, 2021b; Zhang et al., 2022a, 2022b; Lui et al., 2023; Deco and Hugues, 2012; Arazi et al., 2017a, 2017b; Churchland et al., 2010; Wolff et al., 2019, 2021; Tognoli and Kelso, 2014; Shine, 2023; Carhart-Harris et al., 2014). Another dynamic feature concerns the intrinsic neural timescales (INT) which can be calculated by the durations of neural activity changes as in temporal receptive windows (TRW; Hasson et al., 2015) or the autocorrelation window (ACW) (Wolff et al., 2022; Hasson et al., 2015; Raut et al., 2020; Golesorkhi et al., 2021a, 2021b; Honey et al., 2012; Ito et al., 2020; Yeshurun et al., 2021). Additionally, the brain's temporal structure of its neural activity also exhibits scale-free dynamics, where patterns repeat over multiple time scales, contributing to complex neural behavior (He et al., 2010; He, 2011, 2014; Turkheimer et al., 2015, 2022; Raichle, 2009, 2010; Hancock et al., 2022; Pang et al., 2022, 2023; Fagerholm et al., 2023; Tognoli and Kelso, 2014; Dumas et al., 2014; Deco et al., 2015; Tagliazucchi et al., 2012; Vohryzek et al., 2022; Le Bihan, 2020). Yet another dynamic aspect is neural synchronization, where different brain regions coordinate their activity in time

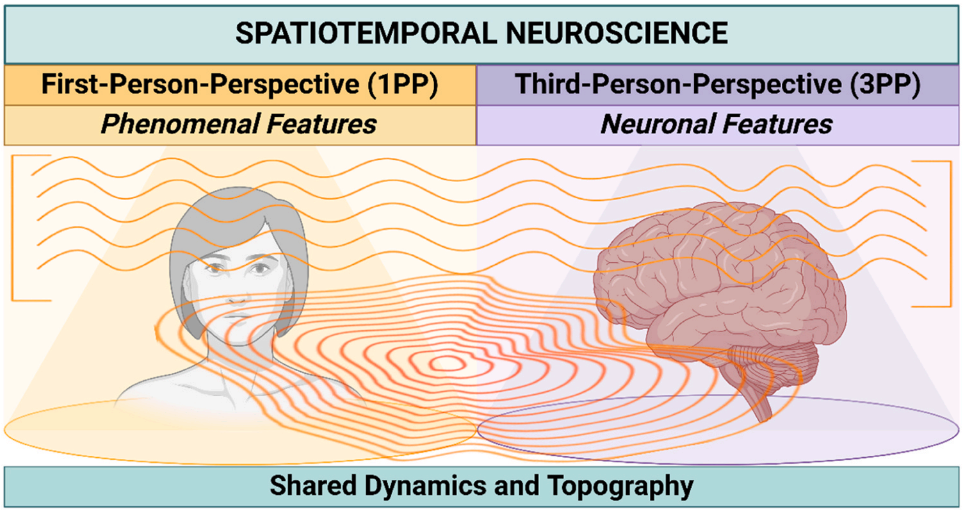

Fig. 1b. The spatiotemporal neuroscience approach, characterized by the shared dynamics and topography between 1PP and 3PP.

(Tognoli and Kelso, 2014; Kelso, 2021a, 2021b; Northoff, 2024; Bolt et al., 2017; Kotz et al., 2018). Finally, more recently, researchers have explored the brain's thermodynamics, including concepts like turbulence and temporal irreversibility, which describe how the brain's activity unfolds over time in ways that do not simply reverse (Deco and Kringelbach, 2020; Deco et al., 2022; Kringelbach et al., 2024; Escrichs et al., 2022). While the intrinsic topographic features of the brain's neural activity concern, for instance, its uni-transmodal topography (Cooper et al., 2022; Margulies et al., 2016; Golesorkhi et al., 2021a) and a three-layered topography of regions related to the self (Qin et al., 2020) (see below for details of these topographic and dynamic features as well as Northoff, 2024).

Together, this brief overview highlights that the brain is characterized by its own intrinsic time and space, reflected in its neural dynamics and topography. Importantly, the brain's inner time and space should be understood within an ecological context, i.e., embedded within, interacting with, and aligning to the broader temporal framework of the external world: this is described as 'temporo-spatial alignment' which, in a nutshell, refers to how the brain's intrinsic dynamics attunes and adapts itself to the extrinsic dynamics of both body (Goheen et al., 2023,

Box 2
– Connecting First- Second- and Third person perspectives - Theories of Consciousness and Social Neuroscience.

We mainly focused on connecting first- and third-person perspective while leaving out the second-person perspective. This box aims to connect all three by first discussing relevant theories of consciousness (first-person perspective connecting to third-person perspective) and, secondly, focusing on social neuroscience as that presupposes the second-person perspective in its connection to the third-person perspective.

Various theories of consciousness have been proposed, all seeking to bridge first-person experience and its phenomenological features (such as qualia) with third-person neural data. (see Northoff and Lamme, 2020 as well as Seth and Bayne, 2022). The Integrated Information Theory (IIT) (Tononi et al., 2016) highlights the key relevance of information integration for consciousness while the Global Neuronal Workspace Theory (Mashour et al., 2020) focuses on the global extension of neuronal activity as triggered or ignited by specific inputs or stimuli. Yet another theory proposed by us is the Temporo-spatial Theory of Consciousness (TTC) (Northoff and Huang, 2017; Northoff and Zilio, 2022) which emphasizes that neural activity and consciousness share their temporo-spatial dynamics as their direct link, that is, their "common currency", by postulating different temporo-spatial mechanisms with one of them being temporo-spatial alignment (see above and Northoff et al., 2023).

How are these (and other) different theories of consciousness related to the spatiotemporal framework as postulated in Spatiotemporal Neuroscience? We suppose that the various theories like IIT, GNWT, and others (like higher-order theory, reentrant theory, or predictive coding) describe an higher-order level, while the temporo-spatial approach, i.e., the TTC, refers to a deeper and more fundamental level (see also Northoff and Lamme, 2020). The TTC assumes that this deeper layer is featured by the brain's temporo-spatial dynamics and shared by the mental level of consciousness. Phenomenological features like qualia (and others) are then intrinsically temporo-spatial dynamical features (see Northoff, 2014a and b) as featured for instance by the temporal features discussed in this paper in the theoretical part. Accordingly, we suppose that the temporo-spatial approach provides a more basic, fundamental and overarching umbrella framework, that is the TTC, for the various theories of consciousness (Northoff and Lamme, 2020).

This addresses the question of linking first- and third-person perspective – but what about the second-person perspective? This is tested for in social neuroscience (see Schilbach et al., 2013; Redcay and Schilbach, 2019; Schilbach and Redcay, 2025, Bolis et al., 2023, Schilbach et al., 2016). We, although tentatively, assume that temporo-spatial dynamics also provides the basis or fundament for the second-person perspective. For instance, one key feature here is the observation of synchrony among the brains of different subjects, which goes along with corresponding synchronization among the subjects' experiences (Schilbach and Redcay, 2025, Scalabrini et al., 2023, 2024). Hence, temporo-spatial dynamics is a most fundamental feature that occurs (and must be 'located') prior to the perspectival distinctions of first-, second-, and third-person perspectives (Northoff and Smith, 2023). These perspectives may thus be more related to us and our methodological strategies and, put more philosophically, reflect our basic epistemological design with its principal limitation to directly experience the brain-mind connection by itself, that is, as such (Northoff, 2018). Consequently, future studies may want to link the degree of neural synchronization between different subjects' brains to the degree of the similarities of their experience of, for instance, their self (Xu et al., 2023).

2024) and environment (Klar et al., 2023; Wolman et al., 2024a and b) (see also Box 2).

From this perspective, the brain's intrinsic temporal dynamics and topography are not isolated but operate as a subsystem, that is, as a part of the larger temporal and spatial structure of the extrinsic world as whole. For instance, while the brain operates within a certain frequency range tied to specific biological and cognitive processes, it remains part of the world's larger temporal ecosystem as whole, influenced by and interacting with the broader rhythms of the environment (like the day-night circle and/or the rhythms of music). Understanding this complex temporo-spatial relationship of brain/body and world requires viewing the brain's intrinsic dynamics and topography as integrated, embedded, and thus nested within the world's more complex and far-reaching spatiotemporal patterns. This emphasizes the need to study the brain in its broader ecological, social, and cultural context: just like the less extended space size of a smaller Russian doll is embedded within the more extended space size of the next larger one, the brain's more limited time-space scales are integrated and thus nested within the much larger time-space ranges of the environment, the world.

2.3. Spatiotemporal approach II – subjective experience: experience exhibits inner time and space

Like the brain, experience can also be characterized by its own inner time and space. The 19th to 20th century philosopher Edmund Husserl expanded on Kant's concept of inner time and space, associating it with consciousness. He specifically applied this notion to its temporal dimension, referring to it as "inner time consciousness" (Fig. 1b; Husserl, 1913, 1928). Husserl and subsequent phenomenologists like Heidegger, Sartre, as well as contemporary philosophers such as Dainton, Zahavi, and Gallagher, suggest that consciousness is characterized by its own inner time and space (Husserl, 1913, 1928; Heidegger, 1927; Sartre, 1943; Zahavi, 2005, 2018; Gallagher, 2005, 2011; Gallagher and Zahavi, 2007; Dainton, 2008, 2010, 2016; Kent and Wittmann, 2021). This perspective challenged the Cartesian view of consciousness as nonspatial and nontemporal, distinct from outer time and space in the natural world (Descartes, 1641).

Husserl emphasized that our consciousness contains continuously changing objects and events, with a succession and continuous temporal flux, akin to what William James described as the "stream of consciousness" (James, 1890). However, this flux is not merely a series of discrete moments; there is also duration and persistence entailing temporal continuity. Single perceptions or cognitions in our consciousness may persist for a period, exhibiting a certain duration, that is, they are, to a certain degree, temporally continuous. For instance, this duration is evident in our perception of events like music. When listening to a melody, we do not just hear each tone as an isolated point in time. Instead, we perceive each tone as having a duration shorter or longer, and our perception of these tones' durations may even extend beyond the duration of the actual physical presence of the tones' sound - Husserl referred to this phenomenon as the "width of present" in our consciousness.

William James elaborated on the temporal nature of consciousness, describing it as a "stream" composed of "substantive parts" (the contents of consciousness) and "transitive parts" (the connections and transitions between contents) (James, 1890). James characterized this temporal structure with two key features: "sensible continuity," where contents persist phenomenally in consciousness after their physical disappearance, and "continuous change," referring to the ongoing flux of different contents in consciousness. These concepts closely align with Husserl's notions of duration and succession, respectively (Northoff, 2014). Interestingly, both duration (sensible continuity) and succession (continuous change) can occur over various timescales. For instance, different contents in our consciousness can persist for shorter or longer durations, and, further, the rate of change between contents can vary from rapid to slow. These varying timescales contribute to the complex temporal structure of conscious experience, including the overlap of present, past, and future moments, as described by Husserl's concepts of protention, retention, and primal presentation (Northoff and Horvat, 2022).

Together, these more philosophical insights into the nature of inner time and space in consciousness, as articulated by Husserl, James, and other phenomenologists (Varela, 1999; Fuchs, 2005; Northoff and Huang, 2017), provide a rich theoretical foundation for understanding the temporal (and also spatial) features of subjective experience. Accordingly, concepts like duration, succession, sensible continuity, and continuous change across various timescales offer a phenomenological framework for exploring the spatiotemporal dynamics of consciousness in First-Person Perspective. This is key for Spatiotemporal Neuroscience: it suggests that subjective experience is characterized and, even stronger, by its temporal and spatial features as these provide the connection to and are shared with those of the brain's neural activity – time-space are the "common currency" of brain and experience (Northoff et al., 2020, 2025). That will be explicated in further detail in the next part when converging Spatiotemporal Neuroscience and Neurophenomenology.

3. Part II: theoretical convergence – connecting Neurophenomenology and Spatiotemporal Neuroscience

3.1. What is "common currency"? Brain and experience share their temporo-spatial dynamics

One key characteristic of Spatiotemporal Neuroscience is that it reframes the question of the connection of subjective experience and neural activity in the terms of time and space: how are the brain's inner time and space related to the inner time and space of experience? Different answers are possible. One may assume that they are different and cannot be related with each other at all. Alternatively, one may assume that brain and experience share, for instance, their temporal dynamics: both may show similar timescales, fluctuations or variability, synchrony and even scale-free dynamics. In that case, brain and experience may share their dynamics and thus their inner time.

The concept of shared temporal dynamics between the brain and experience can be likened to the sharing of language or currency. For instance, even if two people come from different countries and speak different languages, they may still share a common language that allows them to communicate. Similarly, multiple countries, such as those in Europe, share a fiscal currency like the Euro - literally their "common currency." Here, presupposing a figurative sense, we use the term "common currency" to describe the shared internal time and space between the brain and experience (Northoff et al., 2020, 2024; Northoff, 2024).

The assumption of such a common temporal and spatial structure between brain and experience has significant implications for our methodological approach to investigate their relationship. Specifically, the assumption of inner time-space as common currency of brain and experience carries major implications for what we may want to take into view when employing 1PP and 3PP. If brain and experience share temporo-spatial dynamics as their "common currency", 1PP and 3PP may want to target analogous or similar temporal and spatial features. While, at the same time, they need to account for and access them in their distinct manifestation in brain and experience, respectively. Despite the mutual exclusivity of 1PP and 3PP on the purely methodological level, their commonly shared temporal and spatial features for a deeper and more intimate connection of brain and experience. The introduction of the concept of "common currency" may thus allow Neurophenomenology to constrain if not overcome the methodological limitations of 1PP and 3PP investigations, facilitating a deeper and more unified investigation of brain and experience.

Thus, we have discussed so far how Neurophenomenology can benefit from adopting the common currency approach, particularly in

Spatiotemporal Neuroscience. However, this relationship is reciprocal. Spatiotemporal Neuroscience, in turn, can also benefit from applying neurophenomenological rigorous methods that systematically link 1PP and 3PP. Through these approaches, it becomes possible to investigate and identify the shared temporo-spatial dynamics of brain and experience. Accordingly, the convergence of Neurophenomenology, focusing on linking first- and third-person perspective, and Spatiotemporal Neuroscience, postulating systematic brain-experience connection via their common currency, is bidirectional to the benefit of both.

3.2. Measuring temporo-spatial dynamics in both experience and brain I: "non-reductive" and "complex correspondence"

How does such theoretical convergence play out in our phenomenological and empirical investigation? Modern neurophenomenological approaches have developed systematic methods to investigate and quantify phenomenological observations. One such method is experience sampling, which enables the real-time capture of the stream of consciousness through online subjective reports or probes, focusing on standardized features of experience (e.g., meta-awareness, affect, spatial, and temporal perception). This technique allows researchers to measure the evolution of phenomenological invariants over time (e.g., Rodriguez-Larios and Alaerts, 2021; Christoff et al., 2009). Another method is micro-phenomenology, whose detailed interviews provide a deep exploration of the temporal and spatial pre-reflective qualities present in specific experiential episodes. By systematically guiding participants to revisit and describe their lived experiences with high precision and concreteness, micro-phenomenology reveals the underlying structure of these processes (e.g., Przyrembel and Singer, 2018; Petitmengin and Lachaux, 2013; Petitmengin et al., 2013).

How can we relate these phenomenological features of experience to the brain's neural activity? From a purely phenomenological perspective, this may sound absurd as one could seemingly confuse experience and brain or, at best, reduce the former to the latter. This is the moment where Spatiotemporal Neuroscience and its non-reductive method steps in: by assuming that temporal (and spatial) features are shared by neural activity and subjective experience as their "common currency", it must investigate them separately on both neural and mental levels in similar or analogous ways – this amounts to an explicitly non-reductive approach (Northoff, 2014, Northoff et al., 2020, 2025).

By combining these philosophical insights with rigorous empirical methods, researchers might want to investigate how the inner time and space of experience relates to the spatiotemporal dynamics observed in brain activity. This may allow researchers to quantify and model the phenomena that Husserl, James, and others described. In other words, they may provide both phenomenological and neural ways to measure the "width of present," to track the dynamics of "sensible continuity" and "continuous change," and to map the overlapping of past, present, and future in conscious experience. One key feature of Spatiotemporal Neuroscience consists thus in the development of novel measures for quantifying these temporal (and also spatial) features of both brain and experience – for that, it refers to dynamics (in general) and its various measures like variability, scale-free dynamics, timescales, synchronization and thermodynamics as they have been developed in physics and engineering (see above; Northoff, 2024).

The application of these measures to changes over time in both the brain's neural activity and our subjective experience may then demonstrate a certain degree of similarity or correspondence, e.g., "complex correspondence" (Northoff et al., 2025). This will be demonstrated in more detail by various empirical examples in the third part of our paper. Accordingly, this convergence of phenomenology and neuroscience may open new avenues for potentially revealing how the brain's intrinsic spatiotemporal organization is linked to and manifest in a somewhat (complex) corresponding structure of the inner time and space featuring our subjective experience. Such integrative approach could help deepen the investigation of experience/consciousness and its intimate relationship to the brain's neural activity, advancing both Spatiotemporal Neuroscience and Neurophenomenology.

3.3. Measuring temporo-spatial dynamics in both experience and brain II: timescales connect first-person experience and third-person neural activity

In the context of this integrated approach, researchers may aim to identify analogous or similar temporal and spatial characteristics in both 1PP-based experiential reports and 3PP-based neural observations. A pertinent example of this convergence is the exploration of comparable timescales between subjective experiences and neural processes. For instance, Ventura et al. (2024) demonstrated a correspondence between the subjectively experienced attentional width on the one hand and the width of the intrinsic neural timescales (INTs) of brain activity (as measured by the autocorrelation window/ACW) on the other in different meditation techniques. They showed that practices with established phenomenological descriptions of wider attentional focus, like Shoonya meditation, exhibited longer ACW durations compared to narrowly focused practices like Mantra meditation.

This approach thus posits that certain experiential phenomena, such as the duration of specific thoughts or perceptions, which are accessible through 1PP, may correspond to measurable durations in the brain's neural activity observable through 3PP. The common currency hypothesis suggests that these timescales are shared between 1PP experience and 3PP brain activity, thereby providing a unifying framework for connecting spatiotemporal and neurophenomenological approaches. This convergence allows researchers to access the same target, i.e., shared timescales of brain and experience, through both 1PP and 3PP methodologies.

The specification provided by the spatiotemporal common currency approach refines what Neurophenomenology describes as the "disciplined circulation between 1PP and 3PP" (Varela, 1996; Lutz, 2002; Lutz et al., 2024; Lutz and Thompson, 2003; Berkovich-Ohana et al., 2020; Petitmengin and Lachaux, 2013) into a "disciplined iterativity between the timescales as accessed in 1PP and 3PP." Ideally, this would involve an iterative process of refining and adjusting the correspondence between neural and phenomenal timescales through their repeated comparisons of 1PP and 3PP data. This iterative methodology thus extends beyond a disciplined circular process, allowing for progressive refinement of our understanding of the relationship between subjective experience and neural activity. In short, we make a plea for a spatiotemporally-informed Neurophenomenology (see Fig. 2).

4. Part III: empirical convergence of neurophenomenological and spatiotemporal approaches

Having established the theoretical framework for integrating the common currency approach with Neurophenomenology, we can now proceed to demonstrate practical applications of this convergence. We will elucidate how these methodological approaches can be used in conjunction in empirical research, focusing on three distinct mental features: the three-layered nested structure of self, its psychopathology in depression, and its topographic-dynamic reorganization fostered by meditation.

4.1. Self organization: spatial nestedness of the three-layered topography of self in both brain and experience

Background. Self-organization is a fundamental characteristic of complex living systems, defined by their capacity for self-referential interaction with the environment (Maturana and Varela, 1980, 1984). In other words, organisms continuously process information from their surroundings, interpreting it in relation to their own internal state and needs. This principle applies particularly to human experience, where the ability to dynamically integrate information from the environment with one's internal states and needs helps build a stable and coherent

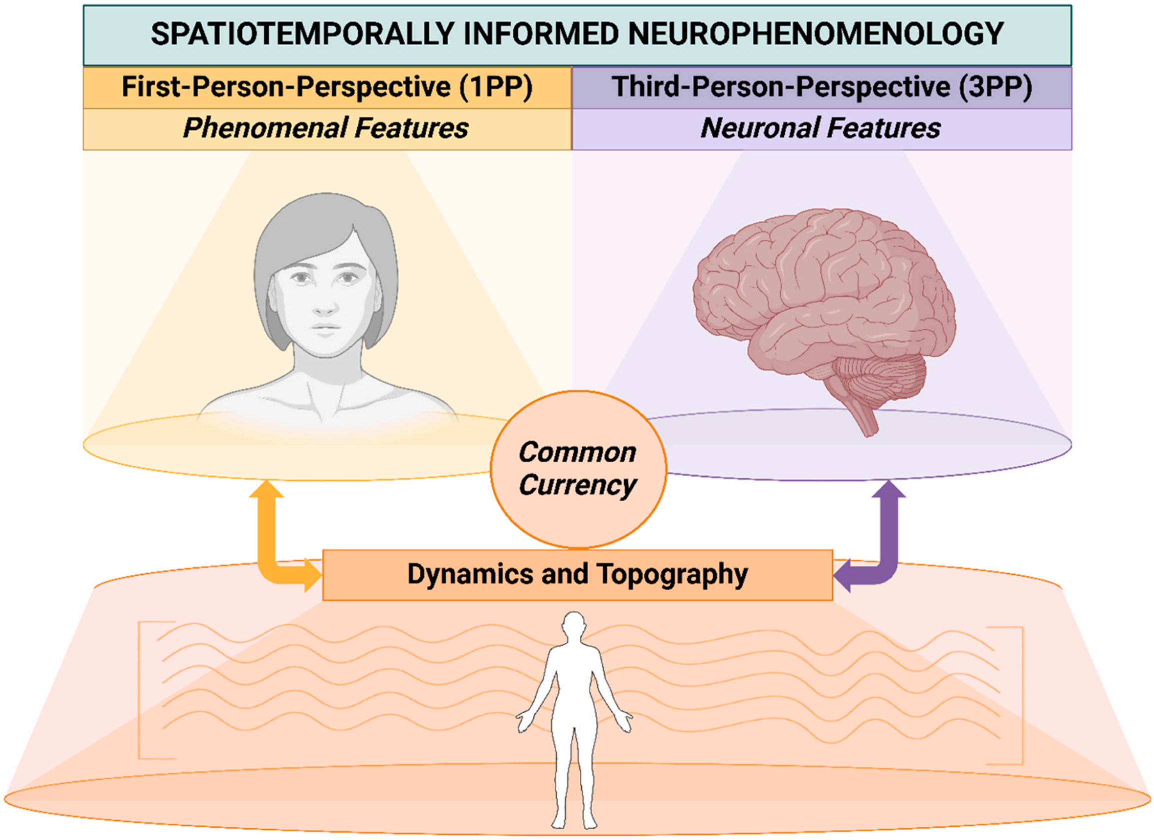

Fig. 2. The spatiotemporally informed neurophenomenological approach, characterized by its common currency, i.e., the intrinsic spatiotemporal features, that is, dynamics and topography that are shared by and connect first-person subjective experience and third-person neural data.

experience of self. Indeed, the self represents the core of our mental life and has been conceptualized in various forms, including mental self, narrative self, autobiographical self, experiential self, bodily self, integrative self, dynamic pattern self, and others (Gallagher and Daly, 2018; Gallagher, 2011; Davey and Harrison, 2022; Klein, 2014; Damasio, 2010; Sui and Humphreys, 2015; Northoff, 2016; Blanke et al., 2015; Blanke and Metzinger, 2009). It also provides a paradigmatic example for comprehending how the spatiotemporal common currency approach can be applied to Neurophenomenology.

Neural three-layered topography of self. Neuroscientific research has contributed to this understanding by investigating the relationship between different layers of the self (each with its own intrinsic dynamics and role in processing self-relevant stimuli) at both the neuronal and experiential levels. This research has uncovered a hierarchical, nested neuronal topography that may represent a fundamental form of self-organization at an experiential level too. Qin et al. (2020) revealed a three-layer topographic organization of the self in the brain, namely intero-exteroceptive, extero-proprioceptive and mental layers, which are reflected in corresponding phenomenological experiences.

The intero-exteroceptive self pertains to inner bodily states, including the sensations of the heartbeat, respiration, and other visceral signals. This represents the most basic layer of self-experience, often occurring below the threshold of conscious awareness but accessible through focused attention, such as during stress when the heartbeat is perceived as racing or during relaxation when warmth is sensed in the stomach. The intero-exteroceptive self provides a fundamental sense of existence and presence within the body. Neurally, it involves subcortical limbic regions, including the thalamus, midbrain, and brainstem, as well as cortical regions like the insula. These areas process interoceptive signals and integrate them with exteroceptive stimuli from the external environment (Garfinkel et al., 2015; Critchley et al., 2004; Wiebking et al., 2010, 2014).

The extero-proprioceptive self relates to the pre-reflective bodily experience, including sensory experience, awareness of bodily boundaries, movement, and spatial orientation in relation to the environment and others. For instance, the perception of limb position and the boundaries of the skin surface contribute to the sense of being an embodied agent interacting with the environment. This layer includes the sense of agency and ownership over bodily actions (Blanke et al., 2015). The neural regions involved in this aspect include those implicated in the interoceptive self, such as the insula, as well as additional areas like the temporoparietal junction (TPJ), premotor cortex (PMC), and anterior medial prefrontal cortex (amPFC). These regions integrate sensory information, thereby maintaining the sense of body ownership, spatial orientation, and interaction with the external world.

Finally, the mental self encompasses reflective self-awareness and the capacity for narrative construction. It involves the ability to reflect on oneself as distinct from others, develop a sense of identity, and create autobiographical narratives (Damasio, 2010; Davey and Harrison, 2022). This is the level of self-experience associated with thinking about one's memories and expectations. When recalling a past event and contemplating its impact on personal identity or planning for the future, the mental self is engaged. It provides the self-referential framework that enables continuity of personal identity over time (Gallagher, 2002, 2011; Gallagher and Daly, 2018). The mental self recruits brain regions associated with both the interoceptive and exteroceptive selves, along with additional areas such as the anterior and posterior cortical midline structures, including the ventromedial prefrontal cortex (vmPFC) and posterior cingulate cortex (PCC) (Northoff et al., 2006; Qin et al., 2020). These structures are involved in self-referential processing, introspection, and narrative construction.

From the spatial nestedness of the brain's three-layered topography of self to the experience of distinct spatially nested layers of self. Together, this hierarchical, three-layer model reveals a nested topographic organization of the self in the brain, where neural regions related to the lower layers are incorporated into the higher layers. Like the Russian nesting dolls, each layer of the self is embedded within the next respective layer and so forth – this reflects that the mental self is not experienced in isolation but against the background of interoceptive and exteroceptive bodily states and their respective self. Conversely, when experiencing the bodily self, it is framed by the context of the mental self.

Phenomenologically, this model illustrates how the self is experienced as a coherent yet multi-layered phenomenon. Each layer —intero-exteroceptive, extero-proprioceptive, and mental— is interwoven within a dynamic temporal-spatial framework that shapes our subjective experience. The interoceptive sensations, bodily awareness, and higher-order reflective processes do not exist separately in our experience; instead, they form a unified sense of self that we continuously perceive. This nested organization captures the complex interplay between different aspects of self-processing and provides a bridge to understanding how these subjective experiences correspond to neural processes in the brain.

Future neurophenomenological studies of self. Future research could employ neurophenomenological techniques where real-time subjective reports of shifting self-experiences (e.g., interoceptive sensations, bodily expression, reflective thought) are synchronized with neural activity using neuroimaging techniques such as EEG or fMRI. To capture the nested hierarchy of selfhood, experience-sampling embedded in immersive virtual reality (VR) environments could measure variations in the experience of the different layers of self, allowing participants to report their experiences in real-time. In this way, neurophenomenological analysis could track how specific neural dynamics align with changes in the subjects' lived experience of their own self and its various layers, possibly offering a mapping of the nested hierarchy of self. This approach could also reveal how disruptions in this hierarchical organization manifest both neurally and experientially. In the following section, we will delve into these disruptions, with a particular focus on the disorganization of self in depression, where recursive thinking, self-polarization, and abnormal slowness emerge as key dynamics Fig. 3.

4.2. Self dis-organization: depression

Background. The previously discussed self-organization has shown to be disrupted in depression, resulting in a fundamental topographic and dynamic disturbance that affects both subjective experience and neural

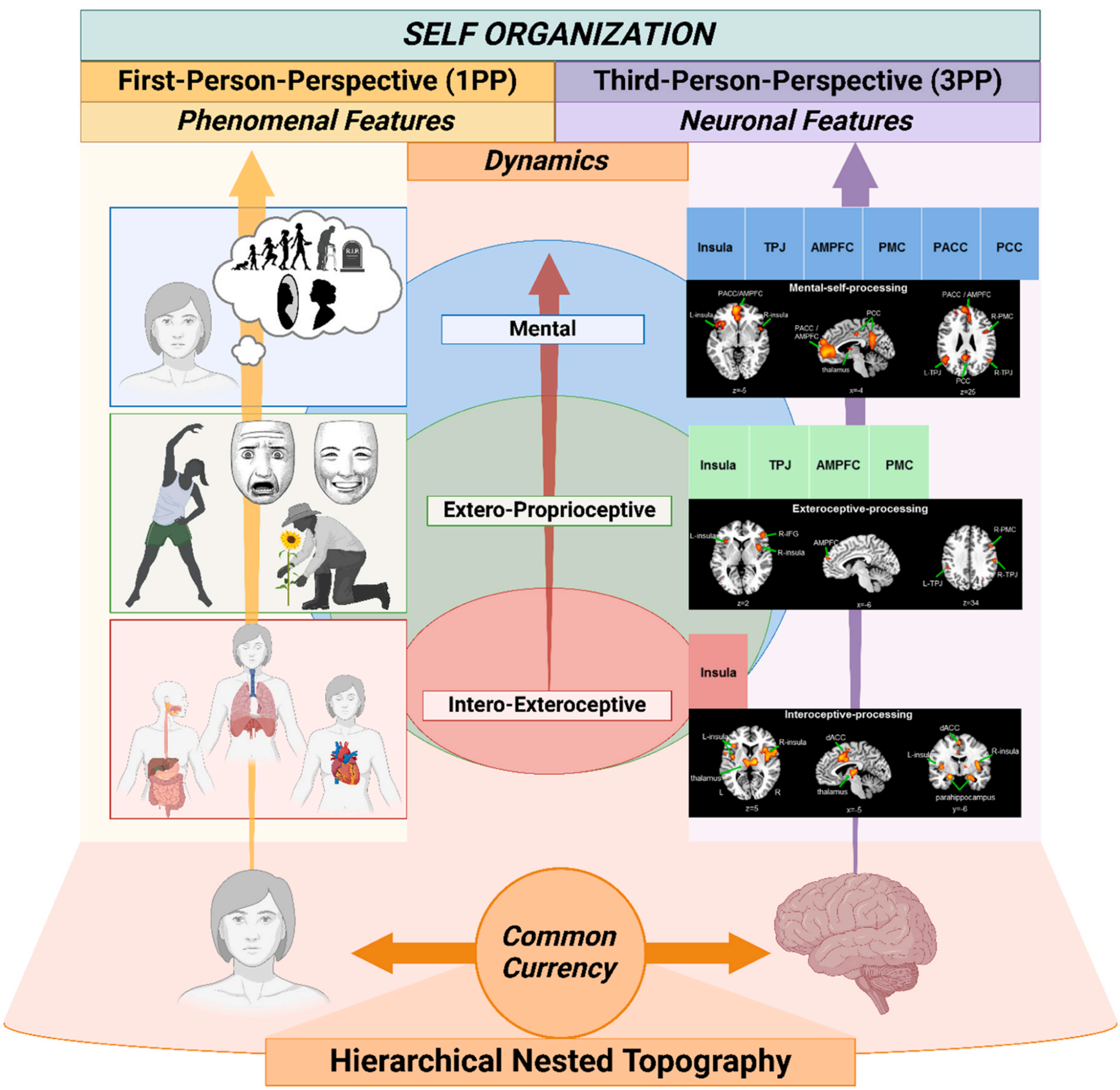

Fig. 3. Schematic representation of the hierarchical spatially nested topography of self-organization. The self is hierarchically organized across three layers: intero-exteroceptive, extero-proprioceptive, and mental. The figure highlights spatial nestedness as common currency that is shared by both brain and experience, that is, representing the spatio-temporal dynamics linking 1PP and 3PP across the three spatially nested layers of self-organization.

activity (Northoff and Hirjak, 2024; Northoff, 2024). In the context of depression, the common currency underlying the disturbance might be observed in three key spatio-temporal dynamics across both 1PP and 3PP: recursion, polarization, and speed.

Recursivity in both brain and experience. Phenomenological literature highlights a disconnection from the body in individuals with depression, where 'the body is turned into a barrier to all impulses directed to the environment' (Fuchs, 2013). Concurrently, there is a shift towards heightened mental processes, resulting in increased rumination and repetitive self-referential thoughts (Scalabrini et al., 2019). This excessive mental activity intensifies withdrawal from embodied experience, creating a recursive cycle that deepens the sense of alienation from both the body and the external world. Such a shift suggests that the hierarchical three-layer topography of the self may become abnormally oriented toward the mental layer, overemphasizing mental self-referential processing while diminishing the processing of bodily and interoceptive aspects of self.

At the neural level, this imbalance aligns with evidence in depression of hyperactivity in brain regions associated with the mental self (Keskin et al., 2023; Scalabrini et al., 2020; Northoff and Hirjak, 2024) and of increased functional connectivity within its brain regions (Greicius et al., 2007; Sheline et al., 2010; Zhou et al., 2010; Hamilton et al., 2011). The overactive recursive thinking, coupled with the hyperactivity and connectivity of the regions of the mental self, may be a manifestation of an intrinsic topographic disorganization underlying the various symptoms in the different domains (Davey and Harrison, 2022; Northoff, 2007). Thus, the disruption of the nested self-organization with abnormal recursive mental activity may be considered a direct link between altered brain dynamics and subjective experience, emphasizing that recursion may be their "common currency".

Polarization in both brain and experience. The topographic imbalance in depression extends beyond mental self-recursion to a polarization in both 1PP experience and 3PP neural observation. Here, individuals with depression often find themselves caught in their own internal world of their self, feeling disconnected from the surrounding environment and from others (Fuchs, 2013). They also report an intense and pervasive self-focus which stands in the fore of their experience. This may be reflected by the neural misbalance in the unimodal-transmodal topography. In depression, the brain's functional gradient is shifted towards the transmodal regions, regions that integrate complex sensory inputs and self-referential processing, at the expense of unimodal regions that process external stimuli. This shift leads to a processing of external stimuli in relation to the self, intensifying self-focus and creating a disruption between the individual with depression and their environment (Northoff, 2007; Keskin et al., 2023). The polarization toward the self can thus represent a clear illustration of how the neural and experiential domains are interlinked through this "common currency."

Abnormal slowness/decreased speed in both brain and experience. The third key dynamic feature linking first-person experiences with third-person observations is the pervasive abnormal slowness of depression. From the 1PP, individuals often describe a slackening of the flow of time and a stagnation of endogenous processes (Stanghellini et al., 2017; Fuchs, 2013). This experience of slowness encompasses a range of mental activities, from thought processes to movements and emotions, contributing to a pervasive sense of retardation (Northoff and Hirjak, 2024; Northoff, 2024; Fuchs, 2013; Stanghellini et al., 2017). Indeed, studies showed that patients with depression exhibit self-reported decreased changes in thoughts (Rostami et al., 2022) and in emotions (van de Leemput et al., 2014). Interestingly, subjects with depression also show reduced speed in their information processing in tasks with time limits (Semkovska et al., 2019).

This experience in individuals with depression has a direct neural counterpart. Studies show that the brain dynamics in depression are abnormally slow, with reduced speed in neural activity across various regions (Guao and Northoff, 2024; Demirtaş et al., 2016). For instance, the visual cortex (i.e., MT+) displays decreased and slower activity, corresponding to a more static perception of moving objects in the environment (Song et al., 2021, 2023; Liu et al., 2022). The motor cortex also exhibits decreased synchronization with global brain activity, which is reflected in the slowed movements typical of psychomotor retardation (Song et al., 2024; Liang et al., 2024). This alignment between experiential and neural levels further supports the notion of a temporo-spatial common currency. The slowing of subjective experience corresponds with decreased speed in neural processing, forming a unified framework for understanding how changes in brain activity are mirrored in the individual's experience of time, movement and perception. Consequently, depression may extend beyond just being a mood disorder but, more basically and fundamentally, it may be speed disorder (Northoff, 2024).

Future neurophenomenological investigation of the self in depression. Future research could deepen our understanding of the basic spatiotemporal dis- or re-organization in depression by integrating neurophenomenological approaches with advanced neuroimaging techniques. For instance, high-density EEG or fMRI could be used to capture key temporal disturbances in depression, such as abnormally slow neural dynamics (e.g., lower ACW or frequency band shifts) alongside disrupted spatial network connectivity, particularly within transmodal regions involved in self-referential thinking. These neural measurements could be paired with moment-to-moment experiential sampling, where individuals with depression rate their sense of mental recursion, polarization toward the self, or subjective slowness during tasks that evoke self-referential processing or bodily disengagement, following recent methodologies in Neurophenomenology (e.g., Christoff et al., 2009; Rostami et al., 2022; Rodriguez-Larios and Alaerts, 2021).

To better capture the subjective dimension of self-disorganization, micro-phenomenological interviews could explore the participants' detailed experiences of recursive mental activity (e.g., "Describe how your thoughts repeat themselves or feel stuck"), perceived polarization toward the self versus the external world (e.g., "How did you experience your connection to your body and the environment?") and slowness (e.g., "How do you experience the flow of time?"). These qualitative descriptions could be systematically coded and correlated with neural measures using advanced computational techniques such as multivariate pattern analysis or functional connectivity mapping. This would allow researchers to directly link the neural manifestations of depressive symptoms, such as increased connectivity in transmodal regions or reduced speed in motor and visual cortices, with subjective reports of slowed time perception and psychomotor retardation.

Additionally, future longitudinal studies could explore whether targeted interventions might help reorganize the hierarchical self-organization in depression. By tracking changes in both temporal (e.g., speeding up of neural dynamics) and spatial (e.g., rebalancing of the transmodal-unimodal gradient) dimensions of brain activity over time, researchers could observe how experiential factors, such as reductions in self-focus or improved bodily awareness, co-evolve with neural reorganization. Therefore, understanding this spatiotemporal link not only provides crucial insights into the dynamics/topography of depressive symptoms and the possibility to develop more targeted diagnostic and prognostic markers, but also lays the foundation for exploring potential pathways toward therapeutic self-reorganization. Interestingly, such reorganization can be observed in practices like meditation, which offers a contrasting model of how the self's dynamics and topography can be reshaped and restored.

Finally, one could also test for more causal relationships. For instance, one could employ measures of speed on both neural and experiential activity. For instance, one could create a timeseries of phenomenological measures like the experience of the timing of specific inputs/stimuli and measure their speed with median frequency (MF) or variability. In parallel, one could apply the same speed measures to neural activity, e.g., MF and neural variability. This, in turn, would allow for not only correlating both phenomenal and neural time series, but also using more causal measures like Granger causality to test for the causal impact of neural speed on phenomenal speed (and vice versa) (see Chuipkta et al., 2025) Fig. 4.

4.3. Self re-organization: meditation

Topographic and dynamic re-organization of experience. Meditation induces both topographical and dynamical reorganization of 3PP brain dynamics and topography and 1PP subjective experience (Cooper et al., 2022; Berkovich-Ohana et al., 2013, 2024; Adventure-Heart and Proeve, 2017; Trautwein et al., 2014; Ray et al., 2021; Hasenkamp et al., 2012; Engen et al., 2018; Jao et al., 2016; Manna et al., 2010; Fox et al., 2014, 2016; Garrison et al., 2015; van Lutterveld et al., 2017b). Unlike the disorganization seen in depression, meditation promotes an experiential shift towards a non-dual mode of being and a more balanced brain functional organization (Cooper et al., 2022; Reddy and Roy, 2019; Osho, 2012). This reorganization reveals how meditation practices can modulate self topography and dynamics, offering a potential pathway to restore coherence and harmony in both 1PP experience and 3PP neural data. In particular, the underlying dynamics of the self-reorganization fostered by meditation are alignment and balance.

Meditators often describe a deepened sense of alignment between their internal bodily states and the external environment (Thompson, 2015; Varela et al., 1991), alongside easier access to interoceptive awareness (Khalsa et al., 2008, 2020; Gibson, 2019). Phenomenologically, this alignment can be understood as an integration with merging of internal and external experiential domains, where the body no longer feels separate from the world but rather becomes a conduit through which sensory and bodily inputs are seamlessly integrated. This grounds the meditator in the present moment, reinforcing a sense of embodied presence and fostering a unified experience of the self as deeply embedded within the surrounding environment. Such alignment suggests that meditators may achieve a state where their body feels not just present but deeply interconnected with the external world.

Topographic and dynamic re-organization of the brain. In a recent

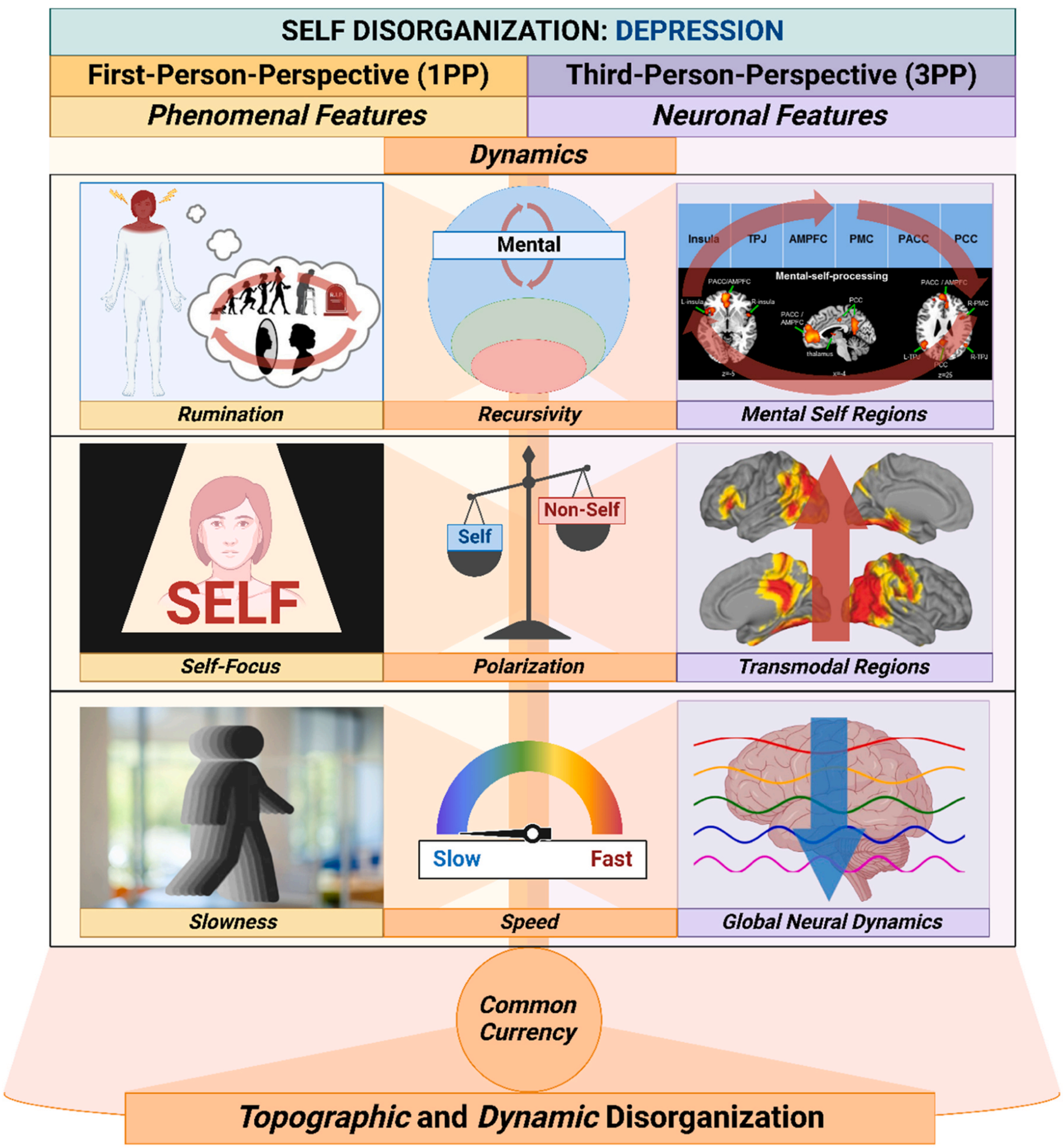

Fig. 4. Illustration of the spatiotemporal dynamics underlying 1PP phenomenal features and 3PP neuronal features in depression. The figure highlights three key dynamics contributing to self-disorganization in depression: recursion/rumination, polarization/increased self-focus, and speed/slowness.

systematic review, Cooper et al. (2022) observed a topographic reorganization of the nested hierarchical structure of self in proficient meditators, highlighting a decrease in activity within cortical midline structures (in particular in the PCC), associated with the mental self, and a relative increase in activity in the subcortical-cortical limbic regions (in particular in the insula), linked to the interoceptive self. Therefore, this reorganized topography fostered by meditation may provide the intrinsic link between what is experienced by proficient meditators and brain activity, reflecting the shift from self-referential, mental processing toward a more embodied experience.

Topographically, this reorganization extends to the spatial interaction of large-scale brain networks. Advanced meditators show a significant shift from the typical anticorrelation to a positive correlation between the DMN, associated with internally oriented self-referential processing, and the CEN, involved in externally-oriented, goal-directed tasks. This change in network connectivity reflects an experiential shift characterized by a blurring of the self-other boundary, aligning with reports of non-dual awareness (Josipovic, 2019, 2021, 2024; Josipovic and Miskovic, 2020), i.e., a state whereby the distinction between self and non-self dissolves. Dynamically, this reconfiguration is further supported by changes in temporal neural dynamics, as demonstrated by Malipeddi et al. (2024). Their study revealed that advanced meditators exhibit a reduced difference in INTs between internally focused and externally focused tasks, suggesting a state of equanimity where attention is balanced and not preferentially tuned to either internal or external stimuli. Notably, the authors highlighted that the smaller was this difference in INTs, the greater was the reported experience of non-duality, indicating a tight link between temporal neural dynamics and the subjective sense of equanimity.

Together, these findings suggest that meditation may induce a spatiotemporal reorganization in the brain, aligning the 1PP and 3PP perspectives through converging their spatial and temporal dynamics. This 'common currency' represents the integration of subjective experience with neural activity, where meditation fosters an experience of

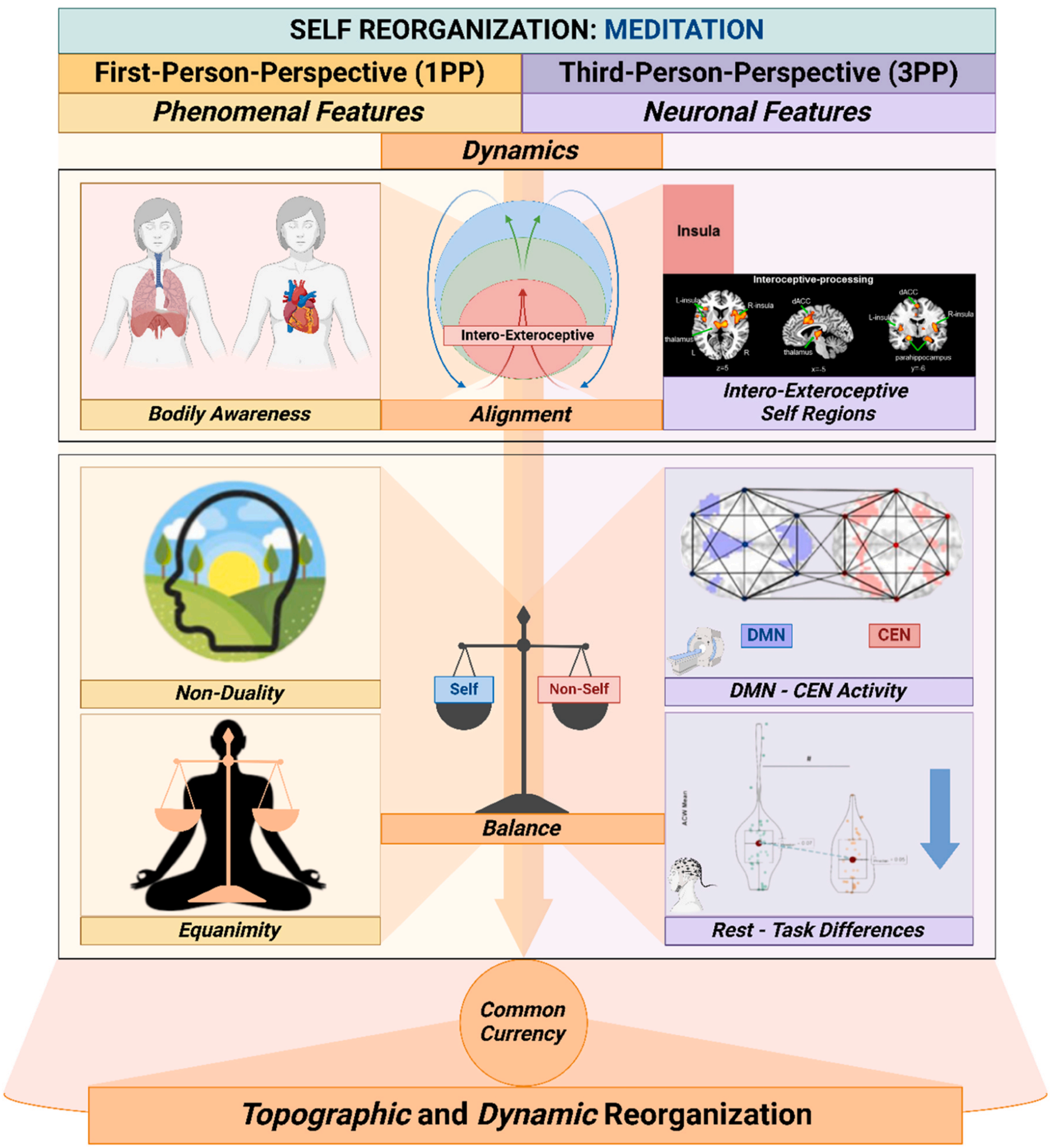

Fig. 5. Illustration of the spatiotemporal dynamics underlying 1PP phenomenal features and 3PP neuronal features in meditation. The figure highlights two key dynamics contributing to self-reorganization in meditation: alignment and balance.

balance between self and environment, both in the neural topography of brain networks and in the temporal dynamics of neural processing. Thus, the practice of meditation seems to facilitate a fundamental reconfiguration that aligns the internal and external worlds, manifesting in an interconnected, non-dual awareness.

Future neurophenomenological investigation of the self in meditation. To further investigate this spatiotemporal link in meditation, future research could combine neurophenomenological and spatiotemporal methods together. For example, researchers could use high-density EEG or fMRI to capture temporal (e.g., through ACW) or spatial (e.g., through network connectivity) neural dynamics. This could be paired with experiential sampling, where meditators use a visual analog scale to rate their sense of self-world boundaries or the width of their attentional focus during meditation, following previous literature (e.g., Rodriguez-Larios and Alaerts, 2021; Christoff et al., 2009; Shoham et al., 2017). Additionally, post-meditation micro-phenomenological interviews could be conducted, using specific questions to elicit detailed descriptions of spatial and temporal aspects of the meditative experience (e.g., "Describe the boundaries of your sense of self" or "How did you experience the flow of time?"). These subjective reports could then be quantitatively coded and correlated with the neuroimaging data using, for instance, multivariate pattern analysis techniques.

Finally, one could also test for more causal relationships. For instance, one could measure the brain's neural dynamics with entropy and variability while, at the same time, also creating and investigating a timeseries of phenomenological reports of equanimity and non-duality during resting state, task states (and eventually also during meditative states) by using continuous behavioral assessment on a screen (see for instance Smith et al., 2022; Chuipka et al., 2025). As in the case of depression, one can then use more causal measures like Granger causality and Transfer Entropy to more directly link neural and phenomenological data beyond their simple correlation Fig. 5.

5. Conclusion

Neurophenomenology provides a methodological toolkit for connecting first-person experience and third-person observation of neural activity in a systematic way. This serves the purpose of overcoming the gap between brain and experience by connecting them with each other. Spatiotemporal Neuroscience shares the same aim but starts from the opposite end, that is, from brain and experience themselves, prior to and independent of our investigation of them and our methodological strategy. Specifically, Spatiotemporal Neuroscience assumes that temporo-spatial dynamics is shared by both brain and experience as their common currency. This raises the question for how such common currency of brain and experience stands in relation to the neurophenomenological strategy of linking first- and third-person perspective.

We here demonstrate how both approaches converge with each other on theoretical and empirical grounds. The commonly shared temporo-spatial dynamics of brain and experience, that is, Spatiotemporal Neuroscience can provide guidance for Neurophenomenology: it can guide the investigator where and what features (like spatiotemporal features) first- and third-person perspective may want to look for when aiming to connect brain and experience. At the same time, the neurophenomenological approach provides systematic methodological strategies for how to employ the "common currency" of brain and experience in our first- and third-person investigation of them. Accordingly, both neurophenomenological and spatiotemporal approaches are complementary and may hugely benefit from each other in their mutual encounter. We thus hope that our article will stipulate future developments for novel methodological strategies to connect brain and experience in a more intimate way through linking their first- and third-person perspectives in both healthy and clinical contexts (e.g., Lutz et al., 2024; Northoff et al., 2020, 2023, 2025).

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